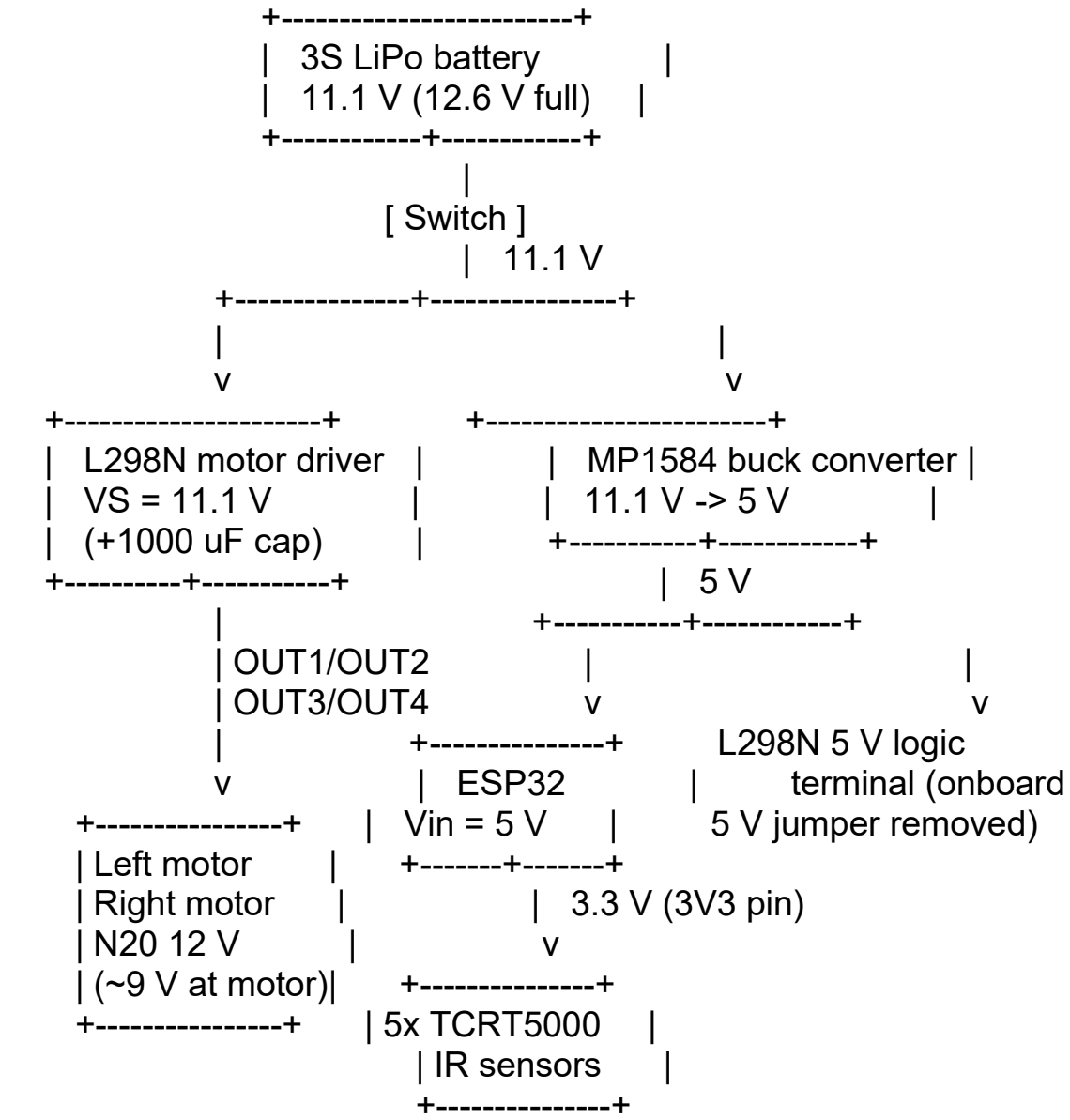
Task 4

POWER BLOCK DIAGRAM (power only; signal wires not shown)



GROUND: battery -, buck, L298N GND, ESP32 GND, sensors and motor driver
all connect to ONE common ground rail.

RAIL SUMMARY
- 11.1 V : battery -> switch -> L298N VS and buck input
- 5 V : buck -> ESP32 Vin and L298N 5 V logic terminal
- 3.3 V : ESP32 3V3 pin -> IR sensors

~9 V : L298N output to motors (about 2 V lost inside the L298N)

Component choice:-

Separate 5 V buck instead of the L298N's 5 V output. The module's onboard regulator is meant for a motor supply of about 12 V or less, and a full 3S pack is 12.6 V. The MP1584 steps 11.1 V down to a clean 5 V for the ESP32 and for the L298N's 5 V logic terminal (the jumper is removed)

#	Item	Qty	Unit ₹	Total ₹
1	ESP32 DevKit V1, 30-pin (ESP32-WROOM-32)	1	500	500
2	MP1584EN mini buck module	1	70	70
3	TCRT5000 reflective IR sensor (bare)	6	20	120
4	Resistor pack, 220 Ω / 10 kΩ, ¼ W	1	100	100
5	L298N dual H-bridge module	1	150	150
6	N20 12 V 300 RPM metal-gear motor	2	350	700
7	N20 motor bracket	2	35	70
8	43 mm rubber wheel for N20 (3 mm D-shaft)	2	50	100
9	Metal ball caster 3/4"	1	60	60
10	3 mm acrylic base plate, laser-cut (~10×12 cm)	1	300	300
11	M3 standoff + screw/nut kit	1	150	150
12	3S 11.1 V 1500 mAh 25C LiPo, XT60	1	1300	1300
13	3S LiPo balance charger (iMAX B3 type)	1	700	700
14	LiPo low-voltage buzzer, 1S–8S	1	100	100
15	XT60 connector pair	2	35	70
16	SPST slide/rocker switch, ≥3 A	1	25	25
1	1000 µF 25 V electrolytic capacitor	2	20	40

17				
18	100 µF 16 V electrolytic capacitor	2	10	20
19	0.1 µF ceramic capacitor	10	2	20
20	Dupont jumper wires M-F, 40 pc	1	70	70
21	Dupont jumper wires M-M, 40 pc	1	70	70
22	22 AWG silicone wire, red + black, 2 m each	1	120	120
23	Perfboard 5×7 cm	2	30	60
24	Male + female 40-pin header strips	1	50	50
25	2-pin screw terminal	4	10	40
26	Micro-USB data cable	1	80	80
27	Zip ties, foam tape, heat shrink	1	100	100
28	Matte white chart sheet + 19 mm black electrical tape	1	150	150
	Parts subtotal			**5,335**

Task 5

Runtime

- Battery: 3S LiPo, 1500 mAh, 11.1 V, which stores 16.65 Wh (11.1 V × 1.5 Ah).
- Usable capacity: 80 %, which is 1200 mAh.
- Typical load: 1200 / 588.6 = **2.04 h (about 122 min)**.
- Worst case with both motors stalled: 1200 / 1773.7 = **0.68 h (about 41 min)**.
- Discharge rate in the worst case: 1.77 A / 1.5 Ah = 1.18C, against a 25C rating (about 37 A).
-
-

Assumptions

- Motor currents (250 mA typical and 800 mA stall each) are estimates, so confirm them on the seller page.
- The ESP32 and L298N logic currents are estimates.
- Buck efficiency is 90 %.
- The runtime figures ignore PWM duty, voltage sag and ageing, so real runtime is likely 10–30 % lower.

Chemistry comparison

Property	3S LiPo (chosen), 1500 mAh 25C	3S Li-ion 18650, 3 × 2600 mAh
Energy density	≈150–200 Wh/kg	≈200–260 Wh/kg
Pack energy	16.65 Wh	28.9 Wh (about 1.7×)
Weight	≈110–130 g, flat pouch	≈140 g cells plus holder
Discharge	25C ≈ 37 A burst	≈1–2C (2.6–5 A)
Cost	≈₹1,300 plus balance charger	≈₹700–900 with holder and protection

Power budget (mA)

Component	Rail	Idle	Typical	Peak
ESP32 DevKit	5 V	50	80	250
5 × TCRT5000 sensors	3.3 V (drawn from 5 V)	47	47	47
L298N logic (V_{SS} + LED)	5 V	30	30	30
5 V rail total	5 V	127	157	327
5 V rail as seen at battery (÷ 0.9 buck efficiency, ÷ 11.1 V)	11.1 V	63.6	78.6	163.7
2 × N20 12 V motors	11.1 V	0	500 (250 each)	1600 (8 each)
L298N motor-supply quiescent	11.1 V	10	10	10
Battery total	11.1 V	**73.6**	**588.6**	**1773.7**